



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

9188/1

PAPER 1 Multiple Choice

NOVEMBER 2012 SESSION

1 hour

Additional materials:

Electronic calculator and/or Mathematical tables

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (Type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 15 printed pages and 1 blank page.

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Data

speed of light in free space,	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space,	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
elementary charge,	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant,	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant,	$u = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron,	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton,	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant,	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant,	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant,	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant,	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall,	$g = 9.81 \text{ m s}^{-2}$

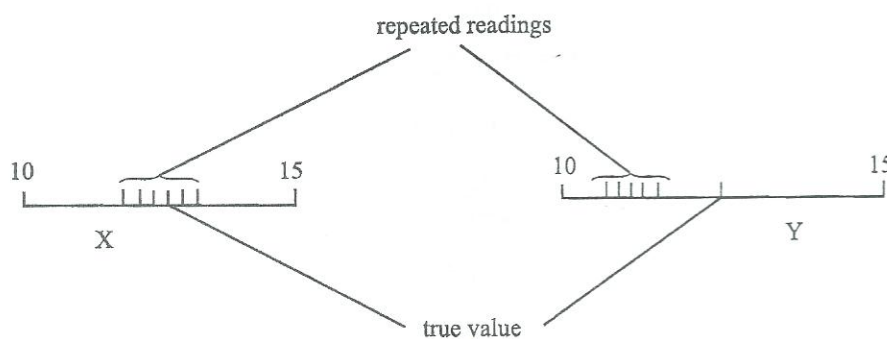
Formulae

uniformly accelerated motion,	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas,	$W = p\Delta V$
gravitational potential,	$\phi = -\frac{Gm}{r}$
refractive index,	$n = \frac{1}{\sin C}$
resistors in series,	$R = R_1 + R_2 + \dots$
resistors in parallel,	$1/R = 1/R_1 + 1/R_2 + \dots$
electric potential,	$V = \frac{Q}{4\pi\epsilon_0 r}$
capacitors in series,	$1/C = 1/C_1 + 1/C_2 + \dots$
capacitors in parallel,	$C = C_1 + C_2 + \dots$
energy of charged capacitor,	$W = \frac{1}{2}QV$
alternating current/voltage,	$x = x_0 \sin \omega t$
hydrostatic pressure,	$p = \rho gh$
pressure of an ideal gas,	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
radioactive decay,	$x = x_0 \exp(-\lambda t)$
decay constant,	$\lambda = \frac{0.693}{t_{1/2}}$
critical density of matter in the Universe,	$\rho_0 = \frac{3H_0^2}{8\pi G}$
equation of continuity,	$Av = \text{constant}$
Bernoulli equation (simplified),	$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$
Stokes' law,	$F = 6\pi\eta r v$
Reynolds' number,	$Re = \frac{\rho v r}{\eta}$
drag force in turbulent flow,	$F = Br^2 \rho v^2$

1 What are the base units of power?

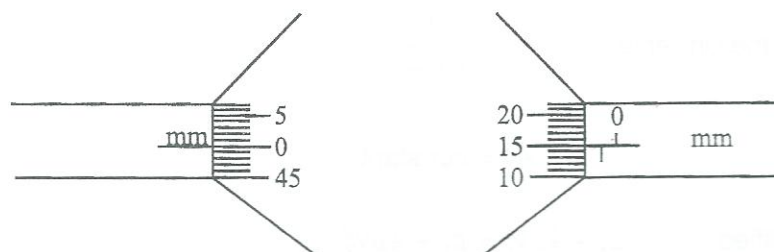
- A Nms^{-1}
- B $\text{kgm}^2\text{s}^{-2}$
- C $\text{kgm}^2\text{s}^{-3}$
- D Js^{-1}

2 The diagram shows repeated readings obtained using instruments X and Y.



The readings from

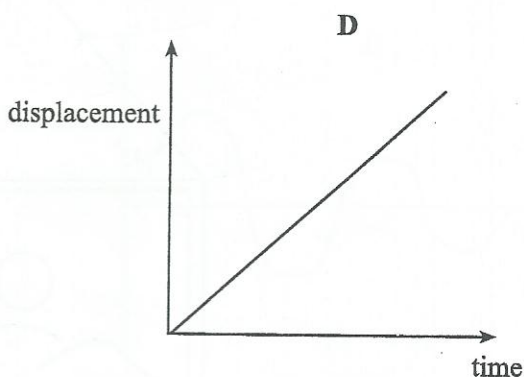
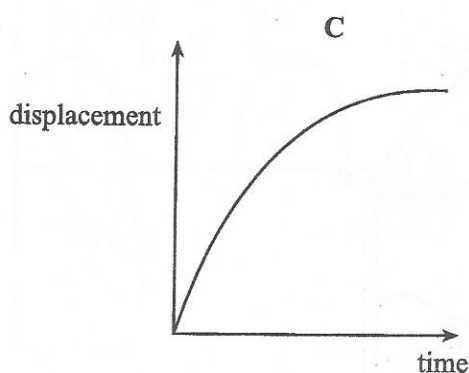
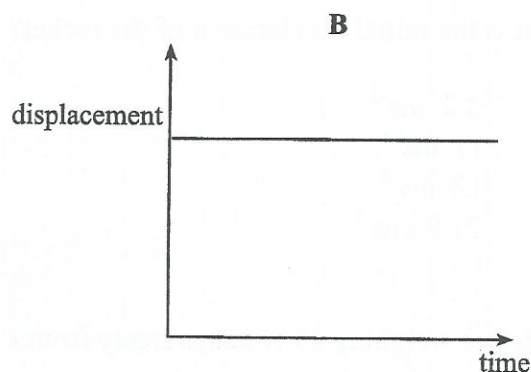
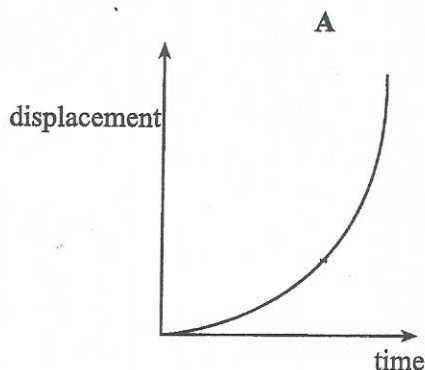
- A X have random errors only.
 - B Y have random errors only.
 - C X have systematic errors only.
 - * D Y have systematic errors only.
- 3 The diagram shows an initial zero reading of a micrometer screw gauge and an observed reading for the diameter of a tin wire.



What is the diameter of the wire?

- A 0.515 mm
- B 0.517 mm
- C 0.65 mm
- D 0.67 mm

- 4 Which graph shows an object moving with increasing velocity?



- 5 During a soccer match, a striker heads a ball at a height of 2 m, giving it a horizontal velocity of 15 ms^{-1} .

What is the speed of the ball as it hits the ground?

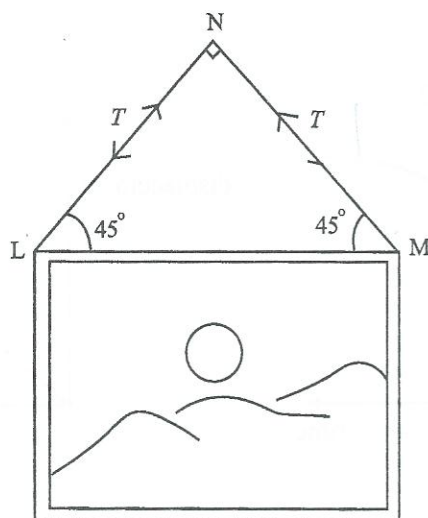
- A 6.3 ms^{-1} .
- B 15.0 ms^{-1} .
- C 16.3 ms^{-1} .
- D 21.3 ms^{-1} .
- 6 When a brick and a feather are dropped simultaneously in free space, from the same height, the
- A feather falls with a higher velocity than the brick.
- B feather and the brick reach the ground at the same time.
- C brick reaches the ground before the feather.
- D the acceleration of the feather is less than that of the brick.

- 7 A rocket which has a lift-off mass of 3.0×10^6 kg develops an initial upthrust of 3.6×10^7 N.

What is the initial acceleration of the rocket?

- A 2.2 ms^{-2}
- B 11 ms^{-2}
- C 12 ms^{-2}
- D 21.9 ms^{-2}

- 8 A picture weighing 15 N hangs freely from a cord LMN as shown.



* What is the tension, T , in each cord?

- A 7.5 N
- B 10.6 N
- C 15.0 N
- D 21.2 N

- 9 The centripetal force for a car moving round a bend is provided by

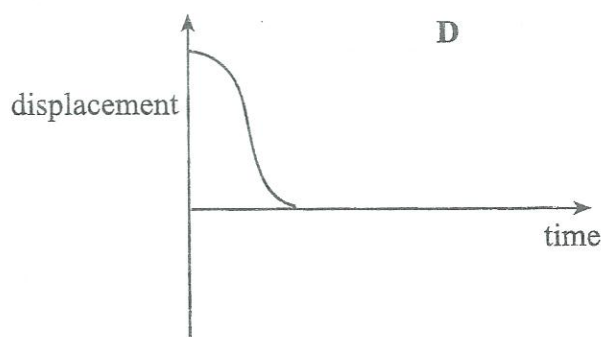
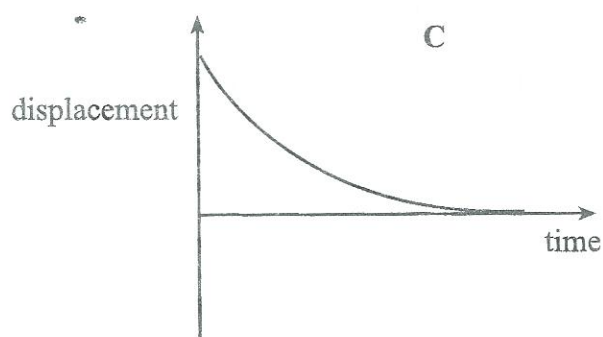
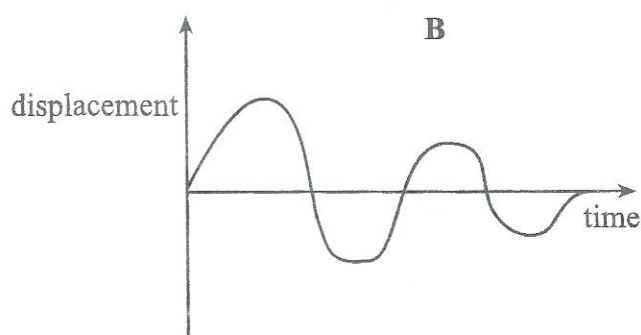
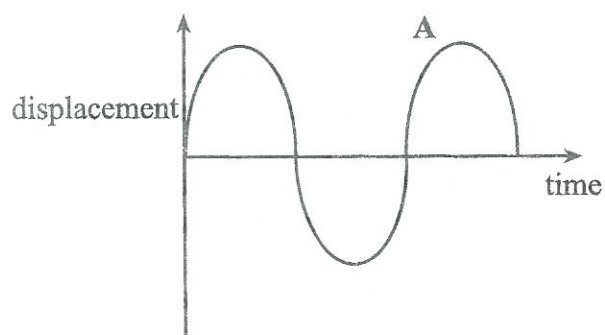
- A its weight and friction.
- B air resistance and friction.
- C friction and mass.
- D air resistance and weight.

- 10 The period of a satellite moving round the Earth in a circular orbit of radius, r , is T .

What would be the period if radius was $r/4$?

- A $T/8$
- B $T/4$
- C $T/2$
- D $4T$

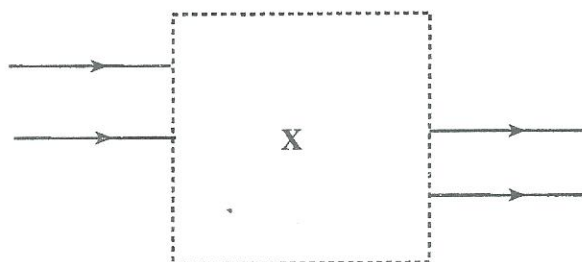
- 11 Which graph represents the behaviour of a comfortable car after going over a hump?



- 12 Which **three** quantities remain constant when a particle executes simple harmonic motion?

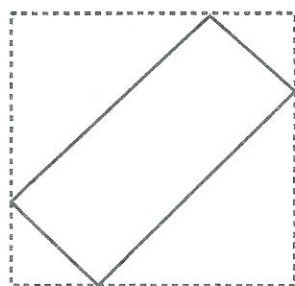
- A acceleration, force, angular velocity
- B total energy, amplitude, angular velocity
- C force, total energy, amplitude
- D amplitude, force, acceleration

- 13 The diagram shows parallel beams of light passing through a piece of optically shaped glass concealed under X.

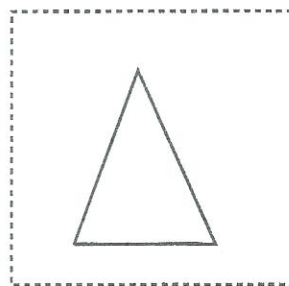


Which shape is under X?

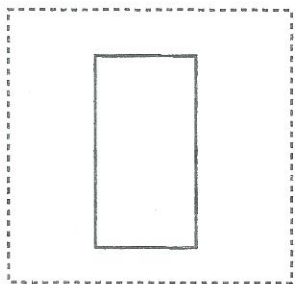
A



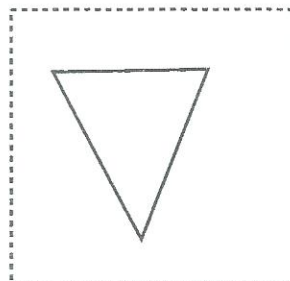
B



C



D



- 14 A wave of wavelength 3 m travelling with speed 12 ms^{-1} in a certain medium enters another medium of refractive index 1.5 times that of the first medium.

Which option shows the correct speed and wavelength of the wave in the second medium?

	wavelength/m	speed/ ms^{-1}
A	2	8
B	3	18
C	4.5	18
D	3	8

- | | | |
|---|-----|----|
| A | 2 | 8 |
| B | 3 | 18 |
| C | 4.5 | 18 |
| D | 3 | 8 |

15 Sound waves **cannot** be

- A refracted.
- B reflected.
- C diffracted.
- D polarised.

16 The electromagnetic wave with the least frequency is

- A X-ray.
- B infrared.
- C Gamma rays.
- D visible light.

17 Which conditions produce a maximum separation of fringes in the Young's double slit interference experiment?

	slit separation	distance of screen from double-slit	wavelength of light
A	small	large	long
B	small	small	short
C	large	large	long
D	large	small	short

18 Which statement is **not** true about stationary waves?

- A Maximum displacement occur on antinodes.
- B The wave profile does not move in the direction of propagation.
- C Energy is transferred along the wave.
- D Points between successive nodes vibrate in phase with one another.

19 The resistance of copper increases with temperature because

- A Electrons move more randomly at higher temperature.
- B Electrons are captured by positive ions at higher temperature.
- C Positive ions impede electrons more at higher temperature.
- D Positive ions are produced in large quantities at higher temperatures.

- 20 Kirchhoff's two laws are statements about conservation of physical quantities.

Which pair correctly represents the law and its respective quantity.

	law 1	law 2
A	energy	charge
B	charge	energy
C	energy	current
D	current	energy

- 21 What is the electronic field strength at the surface of an isolated silver nucleus

$^{108}_{47}\text{Ag}$ of radius r ?

A $\frac{e}{4\pi\epsilon_0 r^2}$

B $\frac{47e}{4\pi\epsilon_0 r^2}$

C $\frac{61e}{4\pi\epsilon_0 r^2}$

D $\frac{108e}{4\pi\epsilon_0 r^2}$

- 22 A $2\mu\text{F}$ capacitor is charged to a potential of 200 V and then isolated. When it is connected in parallel with a second capacitor which is initially uncharged, the common potential becomes 40 V.

The capacitance of the second capacitor is

- A $2\mu\text{F}$.
 B $4\mu\text{F}$.
 C $6\mu\text{F}$.
 D $8\mu\text{F}$.

- 23 A vertical straight conductor of length 0.5 m, carrying a current of 4.0 A, is placed in a uniform horizontal magnetic field of strength 0.1 T.

What angle must the conductor be turned through in the vertical plane so that the force on it is halved?

- A 30°
 B 45°
 C 60°
 D 90°

- 24 An a.c. of r.m.s. value 2 A flowing through a given resistor of resistance, R , dissipates heat at the same rate as a steady direct current, I , flowing through an identical resistor.

What is the current I ?

- A $\frac{\sqrt{2}}{2}$ A
 B $\sqrt{2}$ A
 C 2 A
 D $2\sqrt{2}$ A

- 25 What is the effect of increasing negative feedback on the gain and bandwidth of an operational amplifier?

- A The gain and bandwidth are increased.
 B The gain and bandwidth are reduced.
 C The gain is increased but the bandwidth is reduced.
 D The gain is reduced but the bandwidth is increased.

- 26 Logic gates X and Y have truth tables shown below.

inputs		output
P	Q	R
0	0	0
0	1	0
1	0	0
1	1	1

X

inputs	output
R	S
0	1
1	0

Y

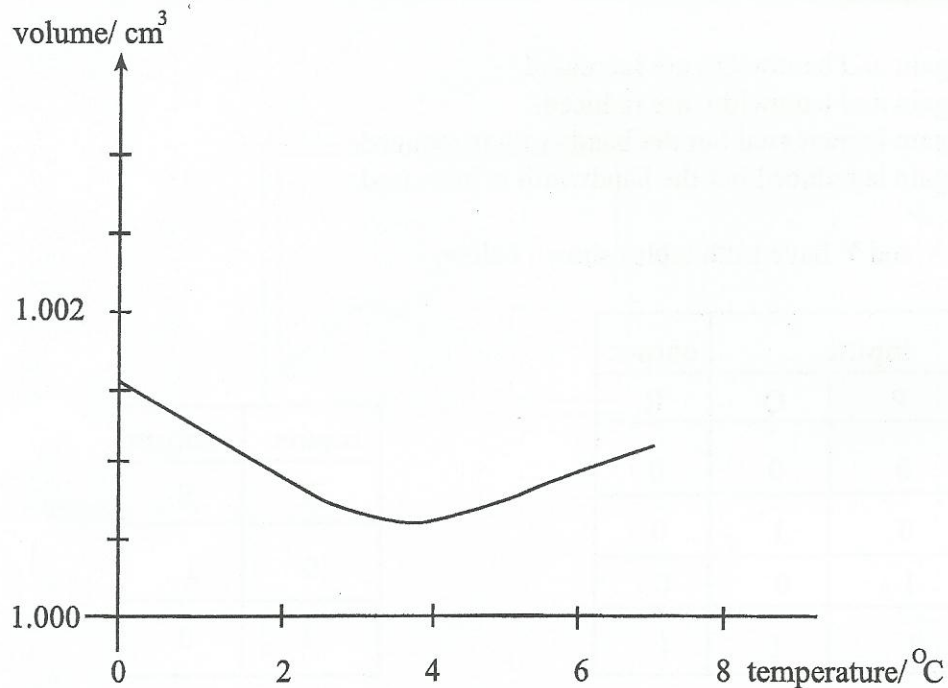
When the output of X is connected to the input of Y, the resulting combination is equivalent to a single

- A NAND gate.
 B AND gate.
 C NOR gate.
 D NOT gate.

- 27 A mass, m , is suspended in turns, on similar wires, X and Y.
If wire, Y, is twice the length and twice the radius of wire X, the ratio $\frac{\text{extension of X}}{\text{extension of Y}}$ is

- A $\frac{1}{4}$.
B $\frac{1}{2}$.
C 2.
D 4.

- 28 The graph shows how the volume of water varies with temperature.



Why would a thermometer using the volume of water as a thermometric property be unsuitable?

- A The decrease in the volume over this range is greater than its increase.
B The variation of volume is non-linear over this range.
C The variation of volume is negligible over this range.
D The variation of volume is not continuous.

- 29 A gas increases its volume from 5 m^3 to 8 m^3 when heated at a constant pressure of 10^5 Pa .

What is the work done by the gas?

- A 300 kJ
- B 500 kJ
- C 800 kJ
- D 3 000 kJ

- 30 Which statement does **not** apply to a platinum resistance thermometer?

- A It can give high accuracy.
- B It measures temperature in small objects.
- C It has a high heat capacity.
- D It can cover a wide range of temperature.

- 31 The r.m.s speed of gas particles with speeds 1, 2, 3 and 4 ms^{-1} is

- A $\sqrt{2.5} \text{ ms}^{-1}$.
- B $\sqrt{7.5} \text{ ms}^{-1}$.
- C $\sqrt{10} \text{ ms}^{-1}$.
- D $\sqrt{30} \text{ ms}^{-1}$.

- 32 The equation $PV = RT$ is for

- A any mass of an ideal gas.
- B one mole of a monoatomic gas.
- C one mole of an ideal gas.
- D any mass of a monoatomic gas.

- 33 In an X-ray tube, an electron of charge, q , is accelerated through a potential difference, V , and then made to strike a metal target.

The minimum wavelength of the emitted radiation is

A $\frac{hq}{cV}$.

B $\frac{qV}{hc}$.

C $\frac{qV}{h}$.

D $\frac{hc}{qV}$.

- 34 Two beams of light, P and Q, of the same wavelength fall upon the same surface causing photo emission of electrons. The photoelectric current produced by, P, is four times that produced by Q.

The ratio $\frac{\text{wave amplitude of beam P}}{\text{wave amplitude of beam Q}}$ is

A $\frac{1}{4}$.

B $\frac{1}{2}$.

C 2.

D 16.

- 35 A photon is

- A a negatively charged particle.
- B a quantum of electromagnetic radiation.
- C an electron emitted from a metal surface by the action of light.
- D an electron emitted from a metal surface by the action of heat.

- 36 Isotopes of any one element have
- A the same chemical properties.
 - B the same numbers of neutron and protons.
 - C different numbers of protons in the nucleus.
 - D the same number of electrons and neutrons.
- 37 The alpha particles are produced when a ${}^7_3\text{Li}$ nucleus is bombarded with certain particles.

What are the bombarding particles?

- A electrons
 - B protons
 - C dentrons
 - D photons
- 38 What deduction can be made from an alpha-particle- scattering experiment?

- A Electrons orbit the atom in well defined paths.
- B The atomic nucleus contains protons and neutrons.
- C The nucleus is very small compared with the size of the atom.
- D There is a great attractive force between electrons and protons.

- 39 ${}^{226}_{88}\text{Ra}$ decays to ${}^{206}_{82}\text{Pb}$ by emission of alpha and beta particles.

What is the total number of beta particles emitted?

- A 4
 - B 6
 - C 14
 - D 20
- 40 The initial activity of a sample of a radioactive isotope containing N_0 nuclei is A_0 .

What is the number of unchanged nuclei when the activity has decreased to $\frac{A_0}{4}$.

- A $0.17N_0$
- B $0.25N_0$
- C $0.69N_0$
- D $0.72N_0$